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Coal vs. renewables: Least-cost optimization of the Indonesian power sector

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ABSTRACT

Indonesia, Southeast Asia's largest economy, ranks globally among the top countries in terms of the envisioned expansion of coal-fired power plants. Despite strongly falling costs for key renewable energies in recent decades, most outstandingly solar PV and wind power, their role in Indonesia's power sector planning is negligible. To assess the potential role of renewable energies in Indonesia's power sector, we develop a cost optimization power sector model. Based on four different scenarios assuming a comparatively slow and a comparatively rapid decrease in the costs for renewable energies, each with and without carbon pricing, we assess capacity expansion, electricity generation, resulting CO_2 emissions, and total system costs until 2040. We compare our results to Indonesia's utility latest power expansion business plans RUPTL, as well as to the country's energy master plan RUEN. We find that official power sector development plans would lead to comparably higher electricity generation costs, due to overcapacity and negligence of future least-cost technologies, especially solar PV. Carbon pricing as low as 5 USD per ton of CO_2 would make coal an economically unviable alternative and foster the integration of biomass and geothermal power plants. In the context of rapidly decreasing costs for renewables, solar PV would be cost competitive with coal even in the absence of carbon pricing by the middle of the 2020–2030 decade. Wind power remains largely uncompetitive across the whole time horizon, due to the lack of substantial wind resources. Our results highlight that Indonesia's power sector planning could be substantially improved, from both a cost and a climate protection perspective.

Introduction

Indonesia is Southeast Asia's largest economy and the 4th most populous country in the world. As a signatory to the Paris Agreement, Indonesia has committed to the goal of limiting the global temperature increase to well below 2 °C above pre-industrial levels and to pursue efforts to limit the increase to 1.5 °C. Achieving this goal will require a carbon-neutral global economy by the second half of the century, for which the global phase-out of coal use without carbon capture and storage in the power sector is a prerequisite (Luderer et al., 2018; Tong et al., 2019).

Strong economic and demographic growth in Indonesia has led to rising electricity demand over the last decade, which has increased at an annual rate of 6% (ESDM, 2021). To date, this demand has been largely covered by fossil fuels, which represent over 90% of the installed power capacities in Indonesia (Enerdata, 2021), while the remainder is provided by hydro and geothermal power plants. Coal plays a dominant role, accounting for more than half of installed power capacities, followed by natural gas, with around one third. With 33 GW of coal capacities under active development, the country ranks 3rd globally after China and India in terms of the future expansion of coal-fired power plants (Global Energy Monitor, 2021). The large expansion of coal-fired power plants in recent years has led to substantially increasing CO_2 emissions from electricity generation, which shows the steepest rise in emissions across all sectors (WRI, 2021).

Recently, global electricity markets have seen a surge in renewable energies (RE), most outstandingly solar PV and wind power, as economically viable and carbon-free electricity sources. Globally, solar PV capacity increased 17-fold from 2010 to 2020, and onshore wind capacity increased fourfold (IRENA, 2021a). The global weighted-average levelized cost of electricity (LCOE) for utility-scale solar PV electricity generation fell by 85%, and by 56% for onshore wind, bringing these RE into a competitive range with fossil fuels (IRENA, 2021b). Despite this global development on the electricity markets, Indonesia has not yet integrated substantial shares of solar PV and wind power. Indonesia's first wind power plant was inaugurated in the Sidenreng

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Rappang Regency in 2018, and its total installed wind capacity is currently around 150 MW. Similarly, its total installed solar PV capacity is around 160 MW, with the first larger utility-scale projects coming online in recent years (ESDM, 2021). Indonesia's latest electricity utility business plan RUPTL 2021–2030, which is Indonesia's most indicative 10-year power sector development plan, foresees the additional development of 14 GW[1] coal power plants in the next decade, while the share of variable renewable energies, solar PV and wind, remains small (PLN, 2021). Indonesian stakeholders report that solar PV and wind projects are non-competitive with coal, in particular due to the prevalence of double-digit interest rates for capital-intensive renewable projects (Ordonez et al., 2021; Ordonez & Eckstein, 2020).

In contrast to the trend in Indonesia, many other countries previously relying strongly on coal, such as China, India or Vietnam, have decidedly integrated variable renewable energies to their power sectors in recent years, thereby reducing the share of coal-fired power plants in their power development plans in favour of renewables (Enerdata, 2021; Global Energy Monitor, 2021). While Indonesia's state-owned utility PLN has recently pledged to strive for carbon neutrality by mid-century, no official targets or plans to achieve this exist (Rahman, 2021). Indonesia's climate target is to reduce greenhouse gases by 29%–41% by 2030 compared to a Business as Usual scenario (see Fig. A1 in the Annex A), while it aims to reach a share of 23% of RE in the power sector by 2025 (Bridle et al., 2018). Most recently, the role of carbon taxes to reach Indonesia's energy and climate targets has been discussed, with the Government of Indonesia reportedly being in the process of drafting an amendment to Indonesia's tax law to incorporate carbon taxes (Indonesian Law Blog, 2021; Rahman, 2021).

In light of the contradiction of relying on coal in the context of plummeting costs of renewables, our study aims at analysing the integration of renewable energy technologies into Indonesia's power sector. This paper investigates cost-optimized pathways to integrating higher shares of RE into the Indonesian power system. It aims to answer the following research questions: 1) Are Indonesia's official power sector development plans cost-efficient? 2) What role could solar PV and wind power play in Indonesia's power sector given their strong cost reductions? 3) What effect would different levels of carbon pricing have on Indonesia's power sector composition? 4) What impacts would the falling costs of RE and the introduction of carbon pricing have on CO_2 emissions and electricity generation costs? By employing a power sector simulation and cost optimization energy model, we systematically explore the different future power sector compositions, its associated CO_2 emissions and electricity generation costs. Technically, we use the Low Emissions Analysis Platform (LEAP, Heaps (2021)), which has also been applied to Indonesia in numerous other studies (DEN, 2019). We examine four different power development scenarios from a cost optimization perspective, as well as two official Indonesian power development plans. We systematically assess capacity expansion, electricity generation, the resulting CO_2 emissions, and total system costs until 2040, and draw conclusions from the comparison.

There are very few scientific studies available that analyse the economic efficiency of Indonesia's power sector. Previous literature examined the drivers for the development of the power sector from a political economy perspective, concluding that coal's cost competitiveness is better able to ensure sustained low electricity prices while supporting PLN's financial health. Coal-related royalties as a source of public finance, the role of coal in regional economic development and the vested interests of key political functionaries owning coal assets are the major drivers of its continued use (Ordonez et al., 2021; Ordonez & Eckstein, 2020). Handayani et al. (2017) modelled different scenarios based on both simulation and optimization in LEAP, without specifically focusing on renewable cost reductions. More recently, some studies have analysed how to reach carbon neutrality using modelling tools (IESR, 2019, 2021a, 2021b, 2021c). Our study aims to help fill this gap by analysing the cost-effectiveness of Indonesia's power sector development under different future assumptions.

This paper is structured as follows. Section 2 describes the rationale behind the considered scenarios. Section 3 presents the modelling methodology. Section 4 shows the results of the future capacity composition in different scenarios, how this compares to policy plans, the role of key technologies, and associated CO_2 emissions and total system costs under each scenario. Section 5 discusses key findings in an integrated perspective with an outlook to policy implications. Section 6 concludes.

Scenario assumptions under falling LCOE

We developed four different scenarios assuming (a) moderate and (b) comparably stronger cost reductions for renewable energies, each (1) with and (2) without carbon pricing (see Table 1). Scenarios (a) and (b) assumed different reductions for both total upfront capital costs and financing costs of electricity generation technologies. RE projects are capital intensive and associated with high upfront expenditures, such as costs of generation equipment, balance of plant components and installation (Steffen, 2020). Thus, total upfront costs, while driven to a large extent by the globally falling costs of key generation components, such as solar PV cells, are also a function of local conditions (IRENA, 2020). Indonesia's ‘local content requirements’, which mandate that 44% of the total value of components in solar PV projects are manufactured locally, further disconnects the country from globally reported cost reductions (GBG Indonesia, 2021). To a large extent, financing costs, which are determined by the interest rates of commercial debt and equity to finance upfront expenditures, reflect local risks associated with the project as a function of regulatory, technical, political, and other risk factors (Feyen & Huertas, 2020). The high upfront expenditures, i.e. the capital intensity of renewable projects, make them particularly sensitive to interest rates (Egli et al., 2019; Egli, 2020). While Indonesian stakeholders report double-digit interest rates for the development of RE projects, typical commercial debt interests as reported by the Internationally Monetary Fund are around 10% per year (IMF, 2022).

Table 1 presents an overview of the assumptions underlying the scenario rationale. The *baseline scenario* assumes moderate cost reductions for solar PV and wind energy costs. These cost reductions could be driven by the globally falling manufacturing costs for main technology components (IRENA, 2020, 2021b), yet hampered by the local content requirements in place. With regard to financing costs, the baseline scenario assumes that these will remain in a double-digit range, reflecting the general country risk and other factors, e.g. constantly changing regulations in the power sector or the reportedly politically-driven aversion of the monopolistic utility towards renewables (Ordonez et al., 2021; Ordonez & Eckstein, 2020). The assumption of prevalent high financing costs is relaxed in the *RE cost reduction scenario*, which features decreasing financing cost rates until 2025 (reaching parity with those of fossil fuels), while capital costs are also assumed to experience a stronger decline, reaching comparable levels to India by 2025 (with figures for India derived from IEA (2020c)). India is taken as the reference country, as the socioeconomics and power sector structure of many of India's utilities are arguably comparable to those in Indonesia (IEA, 2020c). We also examined the impact of CO_2 prices on the electricity sector development for the two previously defined scenarios, that is, in the context of moderate cost reductions for

[1] The total coal-fired capacities under active development are larger than this figure (approx. 33 GW), as they comprise newly announced capacities, as well as capacities in the pre-construction and construction phases (Global Energy Monitor, 2021).

Table 1
Overview of main assumptions for the cost-optimized scenarios (own elaboration).

Scenario name	Rationale	Capital costs for solar PV and wind turbines	Financing cost	Carbon pricing
Baseline	Moderate cost reduction for renewables	Moderate reduction of 3% per year	Prevalent high financing cost rates (10% variable RE, 8% other RE, 7% for fossil technologies)	None
CO_2-price	CO_2 price and moderate cost reduction for renewables	Moderate reduction of 3% per year	Prevalent high financing cost rates (10% variable RE, 8% other RE, 7% for fossil technologies)	USD 30 per ton of CO_2
RE cost reduction	Strong cost reduction for renewables	Decreasing, reaching parity with Indian levels by 2025	Decreasing financing cost rates for all renewables, reaching parity with fossil fuels (7%) in 2025	None
Combined policies	CO_2 price and strong cost reduction for renewables	Decreasing, reaching parity with Indian levels by 2025	Decreasing financing cost rates for all renewables, reaching parity with fossil fuels (7%) in 2025	USD 30 per ton of CO_2

renewables (*baseline scenario*) and in the context of strong cost reductions (*RE cost reduction scenari*o). In the *CO_2 price scenario*, we assumed carbon pricing is introduced in the form a CO_2 tax of USD 30 per ton of CO_2 under the baseline scenario conditions. In the *combined policies* scenario, we likewise assumed the introduction of a CO_2 price of USD 30 per ton of CO_2 but under the RE cost reduction scenario conditions. Accordingly, cost reductions for variable renewable energies are equivalent in the baseline and the CO_2 price scenario. Likewise, the stronger cost reductions for renewables are the same in the RE cost reduction scenario and in the combined policies scenario.

In addition to the four scenarios, our results are compared to the two most recent utilities' power sector capacity development plans by the utility, RUPTL 2019–2028 and RUPTL 2021–2030, (PLN, 2019, 2021), as well as to the capacity expansion envisioned in the long-term energy master plan RUEN by Indonesia's national energy council (DEN, 2017a). RUPTL is the utility business plan and indicative of future installations, while RUEN is a long-term strategy document, which serves as the overarching reference for energy sector planning in Indonesia (Ordonez & Eckstein, 2020). For these power development plans, the future capacity expansion is exogenous to the model and corresponds to the specifications of the plans (RUPTL until 2028 and 2030, RUEN until 2050), while electricity generation, costs and emissions are calculated endogenously based on the lowest operational costs of the installed capacity.

In the cost-optimized modelling environment, the LCOE of a given technology are, under the specified technical constraints, the most relevant factor in determining the optimization objective function (see subsequent methodology section). The LCOE are thus illustrative to understand the decision logic of the model for key technologies and different years and scenarios. Fig. 1 shows the LCOE composition of solar PV and coal for different years and scenarios. Because of high financing cost rates in Indonesia, the share of financing costs is almost 55% for the LCOE of solar PV in 2018. Based on the assumptions presented in Table 1 and Table 2 the LCOE of solar PV would decrease by 18% in 2025 from 2018 levels in the baseline scenario, as well as in the CO_2 price scenario, and by 61%in the RE cost reduction scenario and combined policies scenario for the same year. For 2040, the LCOE of solar PV would experience a 46% reduction in the baseline and CO_2 price scenario, and 73% reduction in the other scenarios. While coal is also assumed to experience a cost reduction over the examined horizon, this technology is mature and cost reductions negligible (approx. 2% reduction between 2018 and 2040). Fig. 1 thus represents coal with uniform LCOE.

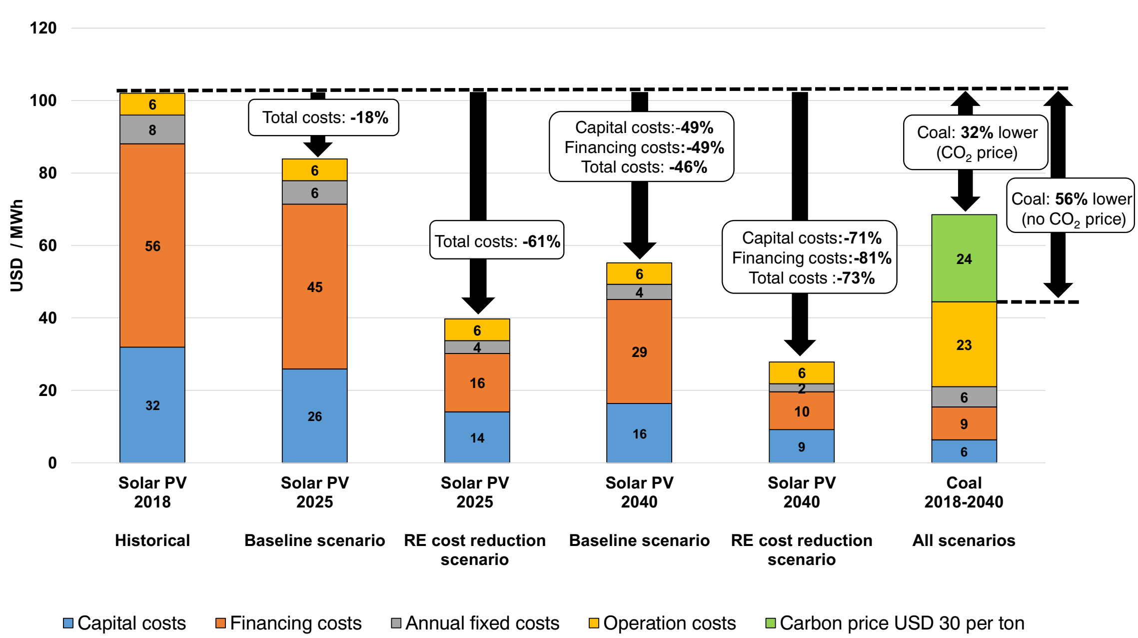

Fig. 1. LCOE of solar PV for the historical year 2018, as well as under the baseline scenario (and the CO_2 price scenario) and RE cost reduction scenario (and the combined policies scenario) for 2025 and 2040, as well as for coal for all scenarios and years. Numbers in the bars indicate costs for the respective component in USD / MWh, while bubbles indicate relative cost reductions or comparison to 2018 solar PV LCOE.

Table 2
Economic and technical parameters of different power generation categories in 2018 and 2040; BAS=Baseline Scenario, RE = RE cost reduction scenario, (source: own compilation based on (A) DEN (2017b), (B) Kalirajan and Syed (2019) (C) IRENA (2020) (D) Cole and Frazier (2019), (E) PricewaterhouseCoopers (2018), (F) ACE (2019), (G) IEA (2020a), (H) BPS (2020), (I) Mone and Hand (2017), (J) DTU (2021), (K) ESMAP (2020).

Power plant type	Capital costs (USD/kW) in 2018/2040	Fixed O&M costs (USD/kW/year) in 2018/2040	Variable O&M costs (USD/MWh) in 2018/2040	Efficiency	Lifetime (years)	Fuel costs (USD/GJ)	Maximum capacity factor
Coal	1400/1340 [A,B]	41.2/39.3 [A,B]	0.12/0.12[A,B]	37% [A,B]	30 [A]	2.4 [G]	80% [A, B]
Open cycle gas turbine (OCGT)	770/700 [A,B]	23/22.5 [A,B]	0 [A,B]	33% [A,B]	25 [A]	6.2[E]	95% [A, B]
Closed cycle gas turbine (CCGT)	750/680 [A,B]	23/22.5 [A,B]	0.13/0.13 [A,B]	56% [A,B]	25 [A]	6.2 [E]	95% [A, B]
Large hydro	2000/2000 [A,B,C,D]	38/34.7 [A,B]	0.65/0.6 [A,B]	–	50 [A]	–	36% [H]
Small hydro	2600/2600 [A]	53/48.8 [A]	0.5/0.46 [A]	–	50 [A]	–	50% [C]
Geothermal	3700/3050 [A,B,C,D]	18/15.9 [A,B]	0.25/0.22 [A,B]	–	30 [A]	–	80% [A, B]
Biomass	1700/1500 [A,B,C,D]	48/40.8 [A,B]	3/2.59[A,B]	31%[A]	25 [A]	1.3 [F]	80% [A, B]
Solar PV	1190/BAS: 610 RE:340[C,D]	11.9/BAS: 6.1 RE: 3.4[C,D]	0 [A,B]	–	25 [A]	–	17% [K]
Wind	1500/BAS: 1210 RE: 1020 [A,B,C,D]	30/26.5 [A,B]	0 [A,B]	–	27 [A]	–	25% [I, J]
Diesel	800/800 [A,B]	8/8 [A,B]	6.4/6.4 [A,B]	45% [A]	25 [A]	15.9 [G]	80% [A, B]
Pumped hydro storage	860/860 [A]	8/8 [A]	1.3/1.3 [A]	80% [A]	35 [A]	–	100%
Battery storage	1570/730 [D]	39.3/18.2 [D]	0 [D]	85% [D]	15 [D]	–	100%

Methodology

This section presents a general overview of the modelling methodology. The model was implemented in the Low Emissions Analysis Platform (LEAP), while the least-cost optimization was computed using the Next Energy Modelling system for Optimization (NEMO), which was directly linked to LEAP (Heaps, 2021; Veysey et al., 2021)[2].

Starting in 2019, the model determines the least-cost capacity expansion and electricity generation for each year until 2040 for the different electricity generation technologies (Table 2). We limit our analysis until 2040 given that the energy system development in next two decades are crucial for the achievement of the goals of the Paris Agreement (Cf. (Luderer et al., 2018)), and the fact that break-even points in the cost-competitiveness of renewables against coal are expected to occur in this timeframe (Fig. 1). Electricity generation is determined for 24 hourly time steps in two seasons (wet and dry) each year. This results in 48 time steps per year, corresponding to 1056 time steps throughout the modelling time horizon.

The modelling is split into five regions (Java-Bali, Sumatra, Sulawesi, Kalimantan, and Papua-Maluku) to account for differences in the load curves, regional renewable capacity limits for each technology, as well as annual and daily RE variations. A reference load curve is taken from Huda et al. (2018) for Java-Bali, and from Winarko et al. (2019) for all other regions. For each region, intraday and seasonal patterns for solar PV and wind power output data are compiled from the Global Solar Atlas (Solargis, 2020) and Renewables Ninja (Staffell & Pfenninger, 2016). Given the lack of interconnection lines between the regions, electricity exchange between regions is not considered.

The growth in annual electricity demand for all scenarios is taken from RUPTL 2019–2028 (PLN, 2019). The growth predicted in RUPTL 2019 (close to 6%) is regarded as a realistic compromise, because it is significantly lower than Indonesia's energy masterplan RUEN (8%), yet higher than the low growth demand projection in the latest business plan of the utility, RUPTL 2021–2030 (4.9%). (KESDM, 2019). Fig. A2 in Annex A presents a visualization and comparison of different demand projections. Additional consumption caused by transmission and distribution losses is implemented as a percentage of electricity demand, and future transmission and distribution losses are based on the projection in the national electricity plan RUKN (KESDM, 2019).

Integrating variable RE into the power system causes additional costs elsewhere in the system (Hirth & Steckel, 2016). These integration costs can be categorized as balancing costs, grid expansion costs, backup costs, full-load hour reduction costs, and overproduction costs (Ueckerdt et al., 2013). Backup costs, full-load hour reduction costs, and overproduction costs are considered endogenously by the model setup. Transmission and balancing costs are specified as additional operational costs. Based on IEA (2018) and Heptonstall and Gross (2020), we integrated these costs with reference values of 6 USD/MWh for solar PV and 4 USD/MWh for wind.

The model also considers battery and pumped hydro storages, which are associated with cost parameters corresponding to the generation technologies. Pumped hydro storage is considered to be off-river storage, for which ample sites exist in all regions of Indonesia (Stocks et al., 2019). For pumped hydro and large-scale battery storage[3], the associated costs are subject to relatively high uncertainty. Large hydro is considered to be hydro power produced using dams, which is constrained in the model to supply electricity in the evening only, while small hydropower plants are assumed to be run-of-the-river plants, implemented without a daily cycle.

In order to calculate the least-cost capacity expansion, the minimum total net present value of all annualized costs of the electricity system over the entire simulated time period is determined by the objective function (Eq. 1).

$$obj = \min \left\{ \sum_y \left[\frac{1}{(1+d)^{y-y_b}} \sum_{PGT} \left[annualized\ capital\ costs + fixed\ costs * capacity_{PGT,y} + operational\ costs * \sum_t generation_{PGT,y,t} \right] \right] \right\} \quad (1)$$

[2] While the aforementioned sources Heaps, 2021; Veysey et al., 2021 provide an in-depth overview of the modelling methodology and this section elaborates the key modelling assumptions and parameters, additional information can be supplied on request.

[3] Of note, the costs for batteries and pumped hydro storage are likewise dependant on the specific capacity costs of each technology (see IRENA, 2017 for a comprehensive overview on costs of electricity storage technologies. Moreover, the values presented in Table 1 are reference values that might vary from source to source.

In which:

$y = year; y_b = base\ year$

$PGT = power\ generation\ technology$

$t = timestep$

$d = discount\ rate$

The net present value of total costs for a given technology was computed based on exogenously determined cost components, classified as upfront capital costs, fixed yearly costs, and operational costs (see Table 2 for an overview of relevant cost assumptions). Conventional operational cost parameters are fuel costs and maintenance costs resulting directly from operation. Carbon pricing costs and variable RE integration costs were also modelled as operational costs. No retirement costs or compensation payments for early retirements are considered. The endogenous model variables are the capacity expansion in each year and electricity generation in each time step, subject to technical constraints on demand and supply balance, availability of power generators, reliable capacity reserve margins, regional renewable capacity limits, and storage constraints.

In the model, the output of a power generation category is constrained only to its available capacity in the respective time step and ramping, i.e. differences between time steps are not limited. This is, however, a technical constraint of real-world applications which also leads to additional costs. These costs are implicitly covered by the model by allocating them to variable RE as integration costs. Incorporating these ramping costs and constraints explicitly into optimization modelling could improve the model in the future once these are available in LEAP.

The model does not represent individual power plants but average patterns of seasonal variations, demand and RE potentials. Therefore, the results should be interpreted as an indication of the cost-optimal dispatch of an average unit rather than a specific single unit. Given the ongoing research and respective uncertainties in the values related to storage facilities, it will be essential to reassess this in the future using updates.

Results

This chapter presents the results of the cost-optimized scenarios. We first discuss capacity expansion capacities in the cost-optimized scenarios up to 2040, the end of the modelling period, and in comparison to Indonesia's latest power sector expansion plans RUPTL (PLN, 2019, 2021). In view of its prominent role in several scenarios, solar PV capacity development is examined in more detail. A sensitivity analysis explored the influence of different CO_2 price levels on power capacity composition in 2040. We then present hourly power generation profiles by scenario in 2040, and compare system costs, resulting CO_2 emissions and carbon revenues.

Capacity

Cost-optimal capacity development up to 2040

Fig. 2 shows the capacity development for all official power development plans considered and the cost-optimized scenarios for selected years. The corresponding values are provided in Table A3 in Annex A.

In 2020, Indonesia had a total installed power capacity of 70 GW, strongly determined by coal-fired power plants (51%, 37 GW), followed by natural gas (21%, 19 GW), and 9% diesel generators (6 GW). Together, fossil fuels accounted for over 90% of installed capacities, with the remainder being mainly hydro (5 GW) and geothermal power plants (2 GW).

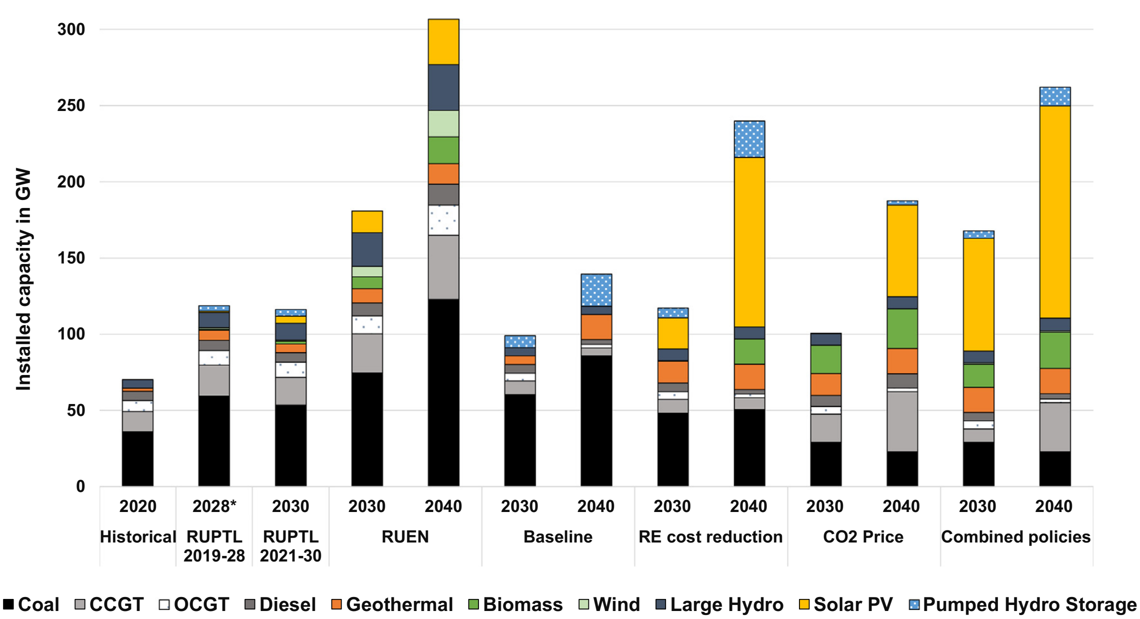

Fig. 2. Installed capacities in 2020, 2030 and 2040 for official power sector plans and cost-optimized scenarios. (*) RUPTL 2019–2028 is shown until 2028, as this is the last year of the plan.

RUPTL 2019 and RUPTL 2021 assume that the strong prevalence of fossil fuels continues up to the end of the decade. In RUPTL 2019, coal is planned to retain its dominant role in the mix, growing to 60 GW by 2028. Natural gas is planned to grow to 30 GW, and diesel generators to remain at 6 GW. Expansion of carbon-free electricity is planned for hydro (10 GW) and geothermal power plants (6.6 GW), while solar PV (1 GW), biomass (0.9 GW) and wind power plants (1.2 GW) are envisioned to play a rather limited role. RUPTL 2021 paints a similar picture to RUPTL 2019 for 2030 values, but reduces coal capacity expansions by 5.7 GW and natural gas by 2.7 GW. Solar PV increases to 4.8 GW and biomass to 1.9 GW, while geothermal capacity declines by 1 GW compared to RUPTL 2019. There is an increase in pumped hydro storage from 3.5 GW in RUPTL 2019 to 4.2 GW in RUPTL 2021.

The energy master plan RUEN envisions a strong expansion of fossil capacities, particularly coal, to an estimated 123 GW by 2040, more than 40 GW beyond the baseline scenario, the most favourable scenario for coal development. We regard this master plan as only broadly indicative, as it is based on a very high electricity demand growth (8% per year compared to around 6% historically) and thus an unrealistically high power capacity expansion.

In the cost-optimized baseline scenario, i.e. assuming a moderate decrease in costs for variable renewable energies and no CO_2 price, coal also experiences a large expansion, reaching 58.6 GW by 2028, which is very similar to RUPTL 2019 (59 GW for RUPTL 2019). Towards 2040, there is another major expansion of coal to 85 GW in the baseline scenario, or an additional 50 GW coal compared to 2020. Geothermal energy plays an equally important role in all cost-optimized scenarios, including the baseline scenario, reaching 16 GW by 2040 and constrained in its upper bound by regional maximum capacity limits. Flexibility is provided by pumped hydro storage (20 GW), while other renewables such as solar PV, wind or biomass remain strictly absent. The share of coal in 2040 increases to 60%, natural gas only comprises 6%, while geothermal increases to 12% of total installed capacities.

In the RE cost reduction scenario, i.e. without a carbon price, but with stronger cost reductions for renewable energies, solar PV reaches 20 GW by 2028, and 111 GW by 2040. Indeed, given the lack of substantial wind potentials, any increase in variable renewable capacities is only due to solar PV in all cost-optimal capacity mixes except the baseline scenario. Coal still plays an important role in the RE cost reduction scenario, reaching around 50 GW by 2030 and remaining at that level until 2040.

In the CO_2 price scenario, i.e. assuming moderate cost reductions for RE but adding a CO_2 price of 30 USD per ton, coal capacities are reduced to 30 GW by 2030 (30 GW less than RUPTL and the baseline scenario) and 20 GW by 2040. Due to its lower carbon content, natural gas is cost-competitive and reaches 14 GW by 2030. In this scenario, no solar PV capacity is built in the 2020–2030 decade, but reaches 60 GW by 2040, suggesting solar PV becomes competitive during the 2030–2040 decade. Biomass would become competitive in the decade 2020–2030, expanding to 26 GW by 2040, but its further development is capped by its regional capacity limits[4].

Combining a carbon price with strongly falling prices for RE technology leads to a rapid expansion of solar PV, which reaches 75 GW by 2030 and 140 GW by 2040. Coal use becomes uncompetitive and is substantially replaced by natural gas and the rapid expansion of solar PV. Biomass and geothermal expansion occur in the 2020–2030 decade, restricted in their development by regional capacity limits.

Comparing the cost-optimized scenarios to RUPTL

Fig. 3 shows the differences in capacities by technology in the cost-optimized scenarios compared to RUPTL 2019 and RUPTL 2021.

Total capacity in RUPTL (121 GW in 2028 for RUPTL 2019 and 119 GW in 2030 for RUPTL 2021 and the RUEN (estimated 159 GW in 2030)) is significantly above the baseline scenario, with 93 GW by 2030, implying a non-optimal use of installed power capacity. As noted in the methodology section, the cost-optimized scenarios are based on demand growth aligned with RUPTL 2019–2028 (PLN, 2019), which ranges between that of RUEN and of RUPTL 2021. This excludes the possibility that overcapacities in RUPTL compared to our baseline scenario are attributable to substantially different demand projections. As discussed in Section 4.4, installing more capacity than required results in higher system costs, as the capital expenditures incurred generate less electricity than cost-optimized capacity utilization.

As noted above, the coal capacity expansion in the cost-optimized baseline scenario is similar to that in RUPTL 2019. The cost-optimized baseline scenario also foresees an expansion of natural gas, diesel, geothermal and hydro, but requires 30 GW less capacity at the end of the decade, which is broken down into 11 GW less gas, 5 GW less geothermal, 4.5 GW less hydro, 3.5 GW less solar PV, but 6.6 GW of additional pumped hydro storage capacities.

In the context of stronger cost reductions for renewables (RE cost reduction scenario), 15 GW of additional solar PV and 5 GW of geothermal are installed until 2028 compared to RUPTL 2019. The RE cost reduction scenario also requires 14 GW less coal in the same year than RUPTL 2019 or the baseline scenario. Likewise, it requires 12 GW less natural gas (CCGT) and 5 GW of pumped hydro storage. Differences to the 2030 values of RUPTL 2021 are smaller, although coal and natural gas are still higher in RUPTL 2021 (by 5 GW and 9 GW, respectively), while the difference to geothermal capacity is higher in compared to this edition of RUPTL. The difference in pumped storage is reduced to 2.2 GW.

With a CO_2 price of 30 USD per ton in the context of moderate cost reductions for renewable energies (CO_2-price scenario), coal becomes economically less viable, and 30 GW less coal capacity is built by 2028 compared to RUPTL 2019 (or compared to the baseline scenario, and 24 GW less when compared to the 2030 values of RUPTL 2021). Likewise, 6 GW less gas are required compared to RUPTL 2019, while the difference is smaller compared to RUPTL 2021 (0.2 GW). Fossil capacities in RUPTL are replaced by 17 GW of biomass power plants and 5 GW of geothermal in the CO_2 price scenario. Given that there are no substantial reductions in the costs of solar PV, this scenario sees no solar PV capacities being built between 2020 and 2030.

Assuming a CO_2price of 30 USD per ton and stronger cost reductions for renewable energies (combined policies scenario) leads to a rapid expansion of RE, with 52 GW of additional solar PV power installed by 2028, as well as 14 GW of biomass and 9.5 GW of geothermal power plants. Coal and natural gas are reduced by 30 GW and 12 GW, respectively, compared to RUPTL 2019. In comparison with the 2030 values of RUPTL 2021, the difference in coal decreases slightly to 24 GW, with additional increases in solar PV (69 GW) and geothermal (11 GW).

Solar PV capacity development

Solar PV could experience the strongest cost reduction across all renewables in the years to come, and consequently plays a key role in all the cost-optimized scenarios. Fig. 4 shows the solar PV capacity additions in 5-year intervals throughout the modelling horizon. In the baseline scenario, no solar PV capacity is deployed. Pricing CO_2 emissions under moderate cost reductions for renewables (CO_2 price scenario) sees no solar PV capacities installed in a cost-optimized power sector until the end of the examined period, i.e. in the timeframe 2036–2040. In this scenario, solar PV costs break even with fossil technologies by 2037, leading to rapid capacity expansion, reaching 60 GW in total in 2040. In the context of rapid cost reductions for renewables but no carbon pricing (RE cost reduction scenario), solar PV already increases until 2035, with a particularly large increase in the last five years of the modelled period. Pricing CO_2 emissions under rapid cost reductions for renewables (combined policies scenario) already leads to rapid

[4] Of note, in our modelling framework, biomass is considered to be managed sustainably thus as of being carbon-neutral. Such a large expansion of biomass power plants would bring along challenges related to land use, land use change and forestry. Annex B provides an overview on limitations within the considered modelling framework.

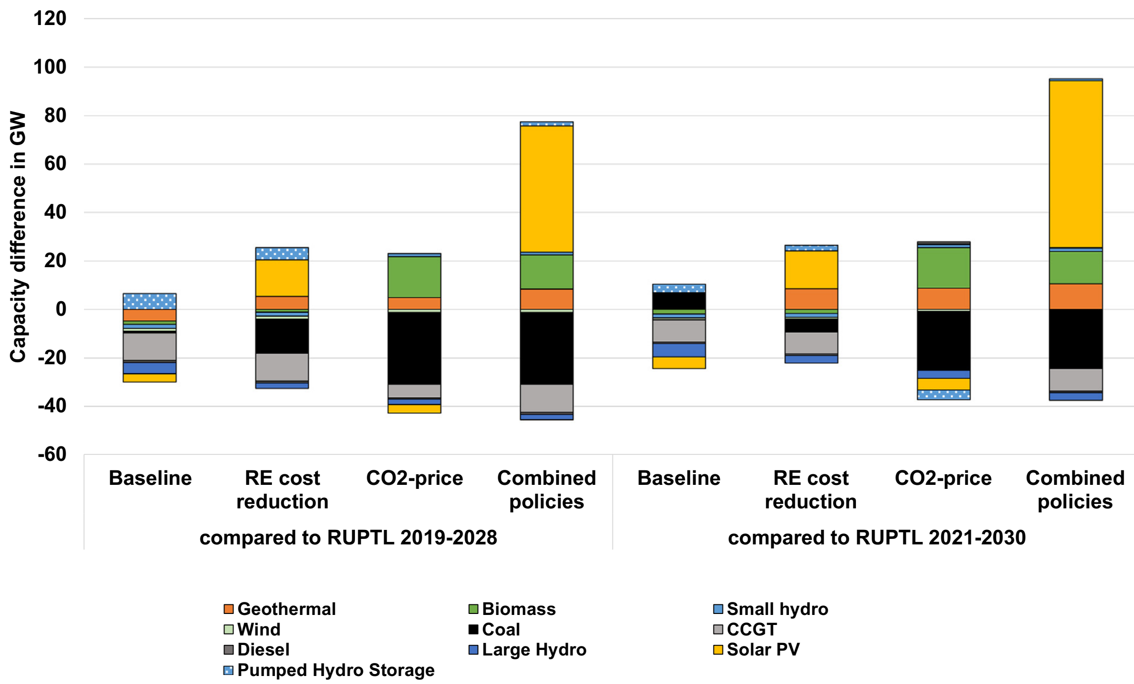

Fig. 3. Capacity differences to RUPTL 2019–2028 for the year 2028 and for RUPTL 2021–2030 for the year 2030 for cost-optimized scenarios.

expansion of solar PV capacities up to 2030, which accelerates even more as costs drop further after 2035.

The role of energy storage

Indonesian stakeholders often address the availability of energy storages as crucial to the development of variable RE (Ordonez & Eckstein, 2020). If no storage capacities are considered, solar PV capacities by 2040 (approx. 110 GW in the RE cost reduction scenario) are reduced by around 20 GW but 90 GW of solar PV would still be installed. Thus, solar PV shows a high potential for economic and technical integration, even without considering energy storages. However, energy storage increases the economic potential of solar PV as it enables the provision of peak power. Based on the underlying cost assumptions (Table 2) in this model, pumped hydro storage is cost-competitive with battery storage throughout the model. Of note, RUPTL 2021 plans an increase in pumped hydro storage compared to RUPTL 2021 (3.5 GW in 2028 of RUPTL 2019 compared to 4.2 GW in 2030 of RUPTL 2021).

The effect of CO_2 emission pricing

Charging USD 30 per ton of CO_2 emissions would trigger a substantial reduction in the use of coal, as presented in the previous section, regardless of the assumptions made concerning RE cost reductions. In order to analyse the effect of carbon prices in more depth, a sensitivity analysis was performed, in the context of moderate cost reductions for renewable energies (by applying a carbon price to the baseline scenario) and in the context of more rapidly decreasing costs for renewable energies (by applying a carbon price to the RE cost reduction scenario). We varied the carbon prices for USD 5, 15, 30, 50 per ton

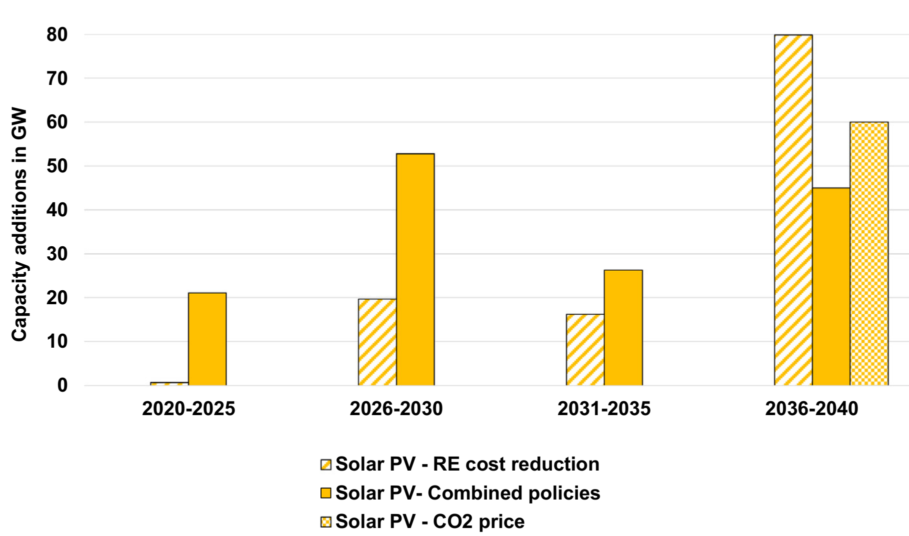

Fig. 4. Development of solar PV capacity in different timeframes for cost-optimized scenarios.

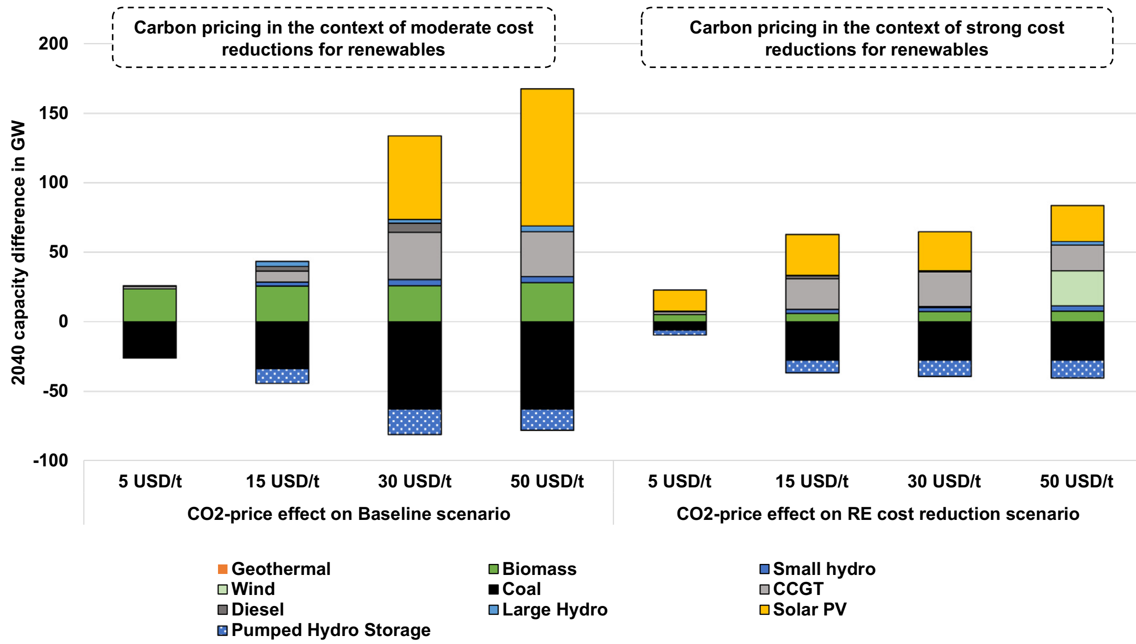

Fig. 5. Capacity differences in 2040 for the baseline scenario (left-hand panel) and the RE cost reduction scenario (right-hand panel) when varying the CO_2 prices in each scenario.

of CO_2 and examined the resulting capacity mix at the end of the modelling horizon.[5] The resulting capacity mix is presented in Fig. 5, which shows the capacity difference in 2040 compared to the 2040 capacity mix of each scenario.

With moderate cost reductions for RE (i.e. basing our sensitivity analysis of carbon prices on the baseline scenario), a carbon price as low as USD 5 per ton of CO_2 reduces the total installed coal capacities by 25 GW by 2040 and adds approximately the same installed capacity of biomass power plants (23 GW) by the same year. Increasing the carbon tax to USD 15 per ton of CO_2 reduces coal by 35 GW, increases biomass by 25 GW, and causes small additions of natural gas and energy storage facilities by 2040. A price of more than USD 30 per ton of CO_2 reduces coal by 63 GW compared to the baseline, and increases solar PV capacities to 60 GW, as well as adding 35 GW of natural gas. Increasing the price to USD 50 per ton of CO_2 additionally expands the share of solar PV to 100 GW.

With stronger cost reductions of renewable energies (i.e. basing our sensitivity analysis of carbon prices on the RE cost reduction scenario), a carbon price of USD 5 per ton of CO_2 decreases coal capacity by 5 GW, although the total installed coal capacities in this scenario are approx. 50 GW by 2040 compared to 85 GW in the baseline scenario. Thus, the resulting coal capacity is around 45 GW, compared to 60 GW in the baseline scenario. This carbon price also triggers the additional instalment of 15 GW of solar PV, from a total of 111 GW to 125 GW in 2040. Increasing the tax to between 15 and 30 USD per ton of CO_2 reduces coal use by around 30 GW to a total installed capacity of 20 GW by 2040. This is equivalent to the capacity mix of the combined policies scenario. Natural gas use then becomes competitive, with an additional 20–25 GW by 2040. A carbon tax of USD 50 per ton of CO_2 triggers the instalment of 25 GW of wind power, highlighting that wind only becomes cost-competitive in the context of strong cost reductions for wind LCOE combined with high carbon taxes.

Daily generation profiles

In view of the large share of solar PV in all cost-optimal scenarios apart from the baseline scenario, we also investigated the potential flexibility challenges of different solar capacity mixes by analysing hourly power generation profiles of a typical day in 2040. We based our analysis on a dry season day in Java-Bali for 2040, because this is projected to have the highest share of solar PV generation of all seasons and regions, and thus the most challenging conditions for balancing supply and demand. Fig. 6 shows the corresponding daily generation profiles.

There is no solar PV in the baseline scenario, and generation and flexibility are provided by pumped-hydro storage or natural gas (Fig. 5, A). In the RE cost reduction and the CO_2-price scenarios (Fig. 5, B and C), maximum solar PV generation occurs between 11:00 and 12:00 if skies are clear, reaching 59 GW in the RE cost reduction scenario and 39 GW in the CO_2-price scenario. Solar PV generation is highest in the combined policies scenario, with a constant supply of 61 GW between 09:00 and 14:00. High amounts of variable solar PV generation result in the need to ramp down other power plants. In the RE cost reduction scenario, as in the combined policies scenario, coal-fired power plants need to be ramped down from 34 GW to close to zero between 06:00 and 09:00, as sunrise initiates solar PV generation, and then ramped up again between 12:00 and 15:00, as solar generation decreases, requiring a very high degree of flexibility. This mode of operation would certainly require refurbishment of coal-fired power plants to provide increased flexibility, as conventional power plants have flexibility limits depending on their age, design and technology[6] (IRENA, 2019). In the CO_2 price scenario, the lower share of solar PV and the comparatively high share of gas compared to the scenarios assuming a stronger decline in cost reductions for RE result in natural gas balancing solar PV generation, without the need to ramp down coal or other technologies.

[5] Under these assumptions, the sensitivity of USD 30 per ton of CO_2 corresponds to the CO_2 price and combined policy scenario, respectively.

[6] These refurbishments might make conventional power plants more expensive, but any such additional costs are not considered in the model.

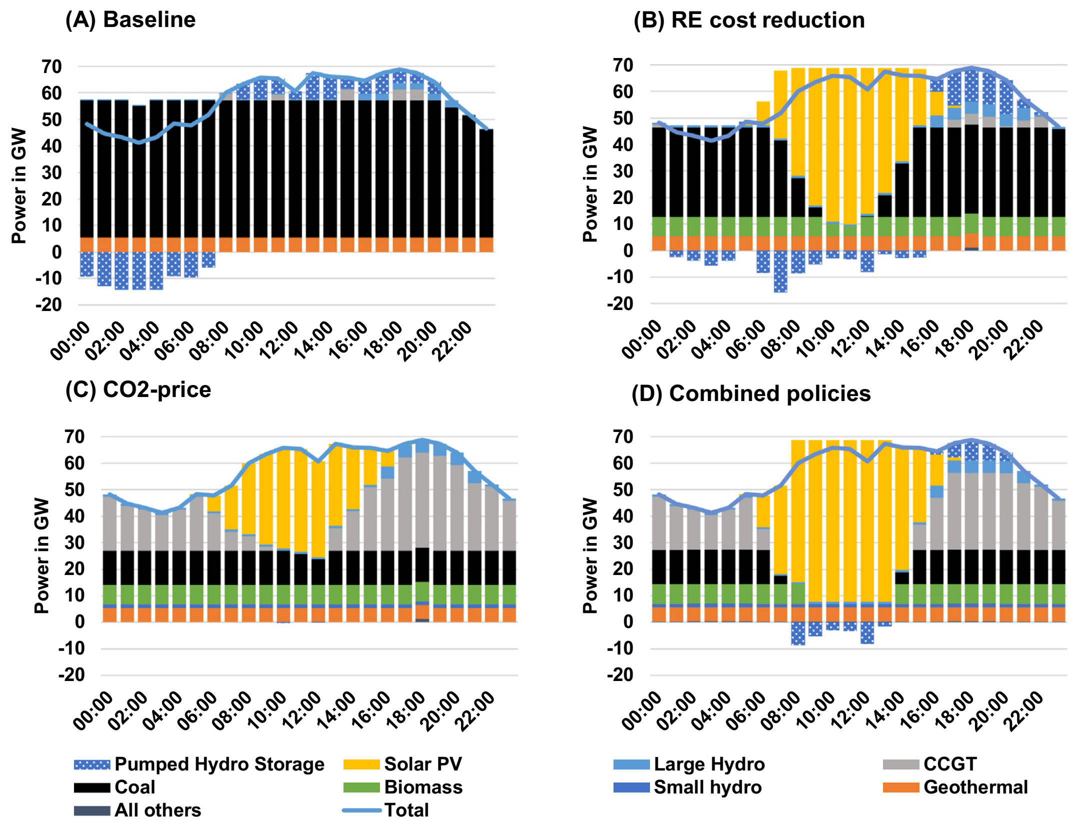

Fig. 6. Daily power generation profiles for different scenarios in Java-Bali in 2040 (dry season).

Electricity-related CO_2 emissions and total system costs

The following paragraphs discuss CO_2 emissions, total electricity production costs as well as the total system costs for the different scenarios. Fig. 7 shows the CO_2 emissions and total electricity production costs in conjunction.

CO_2 emissions

In the context of constantly increasing energy demand, the development of CO_2 emissions reflects the carbon intensity of the resulting electricity mix in a given power sector development scenario. In the baseline scenario, the reliance on carbon-intensive coal causes CO_2 emissions to increase from 180 million tons (MT) in 2019 to 470 MT

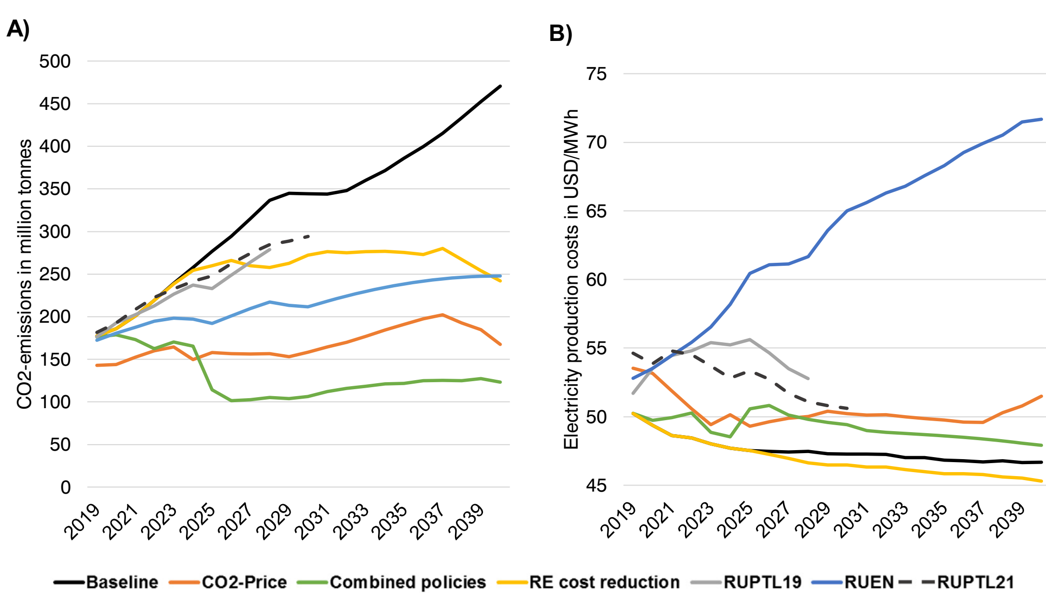

Fig. 7. A) Development of annual CO_2 emissions and B) development of total electricity production costs (carbon pricing costs not included) by scenario and power sector development plans, in the timeframe 2019–2040.

Table 3
Cumulated emissions until 2040 for varying CO_2 prices (under RE cost reduction assumptions).

CO2-price	0 USD/t (RE cost reduction)	5 USD/t	30 USD/t (Combined policies)	50 USD/t	70 USD/t
Cumulated emissions	5.5B T	4.3B T	2.9B T	1.5B T	1.2B T

by 2040 (Fig. 7). The installation of geothermal power plants between 2029 and 2032 reduces the CO_2 intensity of electricity generation and emissions remain constant in this period.

In the context of strong reductions in the costs for renewables (RE cost reduction scenario), Indonesia's power sector still has coal capacity additions, especially in the first half of the 2020–2030 decade, before geothermal and solar PV power plants become competitive and capacity additions of these technologies reduce the CO_2 emission intensity between 2025 and 2030. However, in the 2030–2040 period, the power sector still sees a combination of renewables and coal capacity additions, leading to more or less constantCO_2 emissions. By 2040, the RE cost reduction scenario leads to annual emissions of approx. 240 MT CO_2, which are half the baseline scenario emissions.

Putting a price on CO_2 emissions (CO_2 price scenario) would render coal uncompetitive, and thus decrease emissions from the start of the modelling horizon. Capacity additions in the 2020–2025 timeframe are based on biomass and geothermal power plants in this scenario, which decrease the CO_2 emission factor before natural gas plays a dominant role between 2025 and 2035. Towards the end of the 2030–2040 decade, solar PV breaks even and capacity additions of solar PV rapidly reduce the CO_2 emission intensity. Putting a price on CO_2 leads to emissions of 167 MT of CO_2 by 2040, or 35% of the baseline scenario emissions.

Finally, putting a price on CO_2 in the context of strongly falling costs of renewables (combined policy scenario) leads to the strongest decrease in CO_2 emissions of all the scenarios considered. The carbon tax makes coal uncompetitive, so that solar PV takes a big share of the power sector's composition. This leads to CO_2 emissions of around 120 MT of CO_2 by 2040, or 60% of the 2019 emissions and only 25% of the baseline scenario emissions. RUPTL's last planning year is 2028, but its strong reliance on coal means the CO_2 emission development is aligned with the baseline scenario, reaching 280 MT of CO_2 by 2028, or an increase of 2019 emissions by 50%.

Finally, we computed cumulated CO_2 emissions by the end of the analysed time horizon (2040) resulting from different levels of carbon price building on the RE cost reduction scenario (Table 3). Cumulated emissions decline continuously with increasing CO_2 prices (and are highest in the baseline with 7.1 billion tons (BT)). Notably, emissions are significantly lower for a CO_2 price of 5 USD/t (−1.2 BT) compared to no CO_2price, even though the capacity mix in this scenario is comparable (see Fig. 5).

Electricity production and system costs

Electricity production costs in 2018 USD per MWh of electricity generated a determined by discounting all cost components to the same base year and putting these costs in relation to total electricity generation. While the costs associated with carbon pricing represent a cost block that affects the cost-optimal future capacity mix, carbon pricing also represents a revenue stream for the government that can be recycled in different schemes. Hence, adopting a systemic perspective, we excluded these costs under the assumption that carbon pricing is revenue-neutral (c.f. Indonesian Law Blog (2021)).

Official power expansion plans, above all Indonesia's master plan RUEN, but also the utilities' business plans RUPTL 2019–2028 and RUPTL 2021–2030, lead to the highest system costs and electricity prices of all the scenarios, due to the instalment of substantial overcapacities (Fig. 7). RUEN reaches 70 USD per MWh by 2040. Costs associated with a cost-optimal operation of the capacity mix of RUPTL 2019–2028 are higher than in the optimized scenarios, but still significantly lower than in RUEN, reaching costs of 56 USD per MWh by 2025 and 53 USD per MWh by 2028. After a significant increase in the first half of the 2020–2030 decade, RUPTL 2021 reaches 51 USD per MWh in 2030, close to the CO_2 price scenario. The baseline and the RE cost reduction scenarios, reaching values of USD 45–47 per MWh, have the lowest costs over the time horizon, as their optimization is a function of least-cost technologies, disregarding their carbon context. Accordingly, the CO_2 price and the combined policies scenarios yield higher costs, of USD 52–48 USD per MWh, as cost optimization is a function of the least costs considering emissions. In addition, the costs do not fall continuously because the system costs minimized by the objective function include carbon pricing costs, which are excluded in this figure as discussed above.

To analyse differences in costs over the entire modelling time horizon for the various scenarios and power sector plans, we used cumulated electricity production costs until the end of the time horizon in 2040 (see Table 4). Electricity production costs are presented including and excluding carbon pricing costs. RUEN has the highest cumulated costs at 709 billion USD. The lowest costs are achieved with falling costs for renewables (509 billion USD in the RE cost reduction scenario), while the CO_2 price scenario (550 billion USD) and the combined policies scenario (536 billion USD) are in-between these two. For the scenarios with CO_2 pricing, the difference between the total costs including and excluding CO_2 emission costs represent the carbon revenues. Pricing emissions under moderately falling costs for renewables (CO_2 price scenario) yields 110 billion USD, and 86 billion USD with strongly falling costs (combined policies scenario), ranging between 4 and 6 billion USD annually.

Discussion and policy implications

The results of the cost-optimized power sector modelling show that Indonesia's power sector plans offer scope for both substantial CO_2 emission savings and reducing the overall costs of electricity. While the current official power expansion plans are strongly based on the use of carbon-intensive coal, renewables could play a pivotal role in Indonesia's future power mix, particularly solar PV, but also geothermal and biomass. Reconciling climate protection with sustained low electricity prices is based on a combination of three elements: i) avoiding the installation of unnecessary power plants ii) introducing carbon pricing, and iii) politically fostering cost reductions for RE, in particular for

Table 4
Cumulated system costs throughout the modelling time horizon (2019–2040) excluding and including CO_2 emission costs. The difference represents carbon revenues, the average yearly revenue is computed dividing carbon revenues by 21 years.

	Baseline	RE cost reduction	CO_2-price	Combined policies	RUEN
System costs (excluding CO_2 emission costs)	517B USD	509B USD	550B USD	536B USD	709B USD
System costs (including CO_2 emission costs)	517B USD	509B USD	660B USD	622B USD	709B USD
Difference	0 (No carbon price)	0 (No carbon price)	110B USD	86B USD	0 (No carbon price)
Average yearly carbon revenue	0 (No carbon price)	0 (No carbon price)	5.5B USD	4.3B USD	0 (No carbon price)

solar PV. This section discusses the effects of these elements on the power system's development, as well as policies to achieve cost reductions while reducing CO_2 emissions.

Optimal capacity development

Indonesia's official power expansion plans lead to the highest system costs of all considered scenarios, largely due to the installation of unnecessary power plants. This becomes obvious when comparing the two RUPTLs with the baseline scenario, which has a similar technological composition, but lower capacity requirements and overall electricity costs. Avoiding the installation of unnecessary power plants prevents unnecessary capital expenditures, thus lowering electricity costs. Using realistic electricity demand projections[7] on the one hand, and optimally planning and operating the power plants on the other, is crucial to avoid unnecessarily high electricity costs in the decades to come.

Carbon pricing

Carbon pricing would make the integration of new coal power plants an economically unviable alternative, even at prices as low as USD 5 per ton of CO_2 while fostering low-carbon alternatives. Carbon pricing at USD 30 per ton of CO_2 (carbon price scenario) would also create revenue stream ranging between 4.3 and 5.5 billion USD annually. To put this in perspective, the yearly public revenue collection target from coal and minerals mining imposed on Indonesia's Ministry of Energy and Minerals by the Ministry of Finance is around 40 trillion Indonesian Rupiah, or approx. 3 billion USD (Harsono, 2020). Three quarters of Indonesian coal production and therefore most of this revenue stream is related to coal production for exports. Thus, carbon revenues from domestic carbon taxation in the power sector at USD 30 per ton of CO_2 have the potential to overcompensate the fiscal losses of reduced coal mining for domestic use, while leading to lower emissions and lower electricity production costs than 2019 levels. These revenues could be used for different purposes, including public infrastructure development (Franks et al., 2018; Jakob et al., 2015), social protection (Budolfson et al., 2021), regional economic development of coal-dependent regions (Reitzenstein et al., 2021) or to foster the clean energy transformation (e.g. retrofitting the conventional power fleet for greater flexibility, demand-side management, etc.). In Indonesia's decentralized fiscal system, a part of these carbon revenues could compensate public income losses of coal-dependent regions, such as East- and South Kalimantan, or South Sumatra. Also, increasing electricity prices for end-consumers comes at the possible cost of popularity losses of elected officials, and reducing fuel and electricity subsidies has proven to be represent a major challenge in the past (Inchauste & Victor, 2017; Ordonez et al., 2021). Carbon revenues could thus also be directed back to citizens, in particular towards low-income segments of society, thereby avoiding a trade-off with poverty alleviation and increasing the acceptability of important political constituencies (Klenert et al., 2018; Maestre-Andrés et al., 2019).

Fostering cost reductions for renewable energies

Solar PV could become economically viable in Indonesia by the middle of the 2020–2030 decade, if costs fell quickly (RE cost reduction scenario). Support schemes for RE, such as feed-in tariffs[8] or quotas, have proven to foster the integration of RE technologies in a variety of contexts, but it remains unclear to what extent such subsidies reduce the LCOE of RE technologies (Ragwitz & Steinhilber, 2014).

Decreasing the LCOE of RE technologies can only be achieved by reducing its cost components: either lowering technology and installation costs, or financing costs. Technology costs for solar PV have shown a sharp decrease in recent decades, a decrease that is explained by the well-documented relationship between the cumulative installed volumes of a given technology and its overall manufacturing costs (IRENA, 2020). The global market of solar PV manufacturing is dominated by China, followed by Malaysia, Vietnam, Taiwan and Korea, while Indonesia plays a negligible role in the production of solar PV equipment (IEA, 2020b). Arguably, prevalent local content requirements in Indonesia, which require solar PV projects to have a 44% share of locally manufactured components, cut the country off from the continuous reduction in technology costs elsewhere. Even more, in absence of an Indonesian local manufacturing solar PV industry, such local content requirements might represent a prohibitive barrier to develop solar PV projects. On the other hand, the establishment of a local solar PV manufacturing industry requires both a policy mix with technology-push and demand-pull policies, as well as the existence of related industrial capabilities, a favourable investment context, among other (Hayashi, 2020). Thus, either abolishing local content requirements to enable reductions in solar PV equipment costs by imports, or developing a comprehensive strategy to foster the establishment of a cost-competitive local manufacturing capabilities, will be key to lowering solar PV technology costs in the Indonesian context.

Financing costs are the other large LCOE cost component of capital-intensive renewable energies. Lowering these costs, which are reflected in the weighted average cost of equity and debt, requires a reduction in regulatory uncertainty and thus the risk premiums on capital to finance such projects. Specific project risks are often constituted in the numerous administrative, regulatory, economic, political and other barriers that project developers face when developing a project (Woodman et al., 2017). De-risking RE projects thus requires guaranteeing a stable regulatory environment and reducing administrative and regulative barriers is key to lower financing costs (Schmidt, 2014; Waissbein et al., 2013). Reforming the monopolistic structure of the power market, which in Indonesia has a strong bias towards favouring coal-fired power capacities over RE, might also facilitate RE integration. Indeed, experiences from multiple countries show that competitive markets have replaced coal with cheaper alternatives, i.e. renewable energies and natural gas in recent years (Jakob & Steckel, 2022). However, such far-reaching reforms represent a major institutional change that is hard to achieve in developing countries and emerging economies, given their weaker legal, economic and political institutions (Victor & Heller, 2007).

In Indonesia's non-competitive, monopolistic power market, both falling costs of renewables and a carbon tax might not automatically translate into a high share of low-carbon technologies (Bauer et al., 2021; Eckstein, Nascimiento, et al., 2020). Electricity procurement by the state-owned utility PLN by developing domestic power plants or by entering into long-term power purchase agreements with independent power producers implies that careful consideration of least-cost technologies is pivotal in order to avoid a long-term lock-in into high electricity costs. A sub-optimal choice of technologies ultimately implies a lock-in to either higher electricity prices for end consumers, or higher electricity subsidies.

Conclusions

In this paper, we developed a cost-optimized power sector model for Indonesia and examined the cost-optimal power sector composition until 2040 in the context of moderate and strongly falling costs for RE technologies, with and without carbon pricing. We thereby analysed the role of different technologies in the electricity supply, their associated costs and CO_2 emissions, and compared our results to those obtained by using the power development plans RUPTL 2019 and

[7] Electricity and energy modelling reports in Indonesia and ASEAN neighbors often rely on overall optimistic economic growth and corresponding electricity demand growth projections (Cf. ACE, 2015, 2017, 2020; DEN, 2016, 2019).

[8] Of note, the resulting capacity expansion of the *RE cost reduction* scenario could also be attained by integrating renewable support schemes and thus subsidizing the additional costs of renewables. Yet, this would not yield the system cost reductions as presented in the previous chapter.

2021 of Indonesia's utility and the energy masterplan RUEN by the National Energy Council.

Our findings show that Indonesia's power sector plans offer scope for improvement. Indonesia's power sector development could reconcile climate mitigation with reducing the overall costs of electricity. In particular, we found that avoiding overcapacities, implementing carbon pricing, and fostering cost reductions for solar PV, are of major importance in tapping this potential. Combining these elements is crucial to achieve a rapid reduction in CO_2 emissions while sustaining low electricity generation costs, thereby aligning Indonesia's power sector development with the goals of the Paris Agreement.

Our study highlights that some main objectives behind the continued use of coal-fired power plants, i.e. keeping electricity prices low and collecting public revenue from coal royalties, could still be achieved by using low-carbon technologies and introducing carbon pricing.

The experience of other countries which have installed substantial shares of solar PV in the last decade, such as India and Vietnam, demonstrates how countries with coal endowments, a comparable socio-economic development level, and similar power sector structures can overcome prevalent barriers and substantially exploit their vast solar PV potentials. Against this background, our results highlight how climate protection in Indonesia might not represent a trade-off with economic development or poverty alleviation, but could provide a clear opportunity to reduce electricity costs for industry, commerce and households alike.

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Declaration of competing interest

The authors declare no conflict of interest.

Appendix A

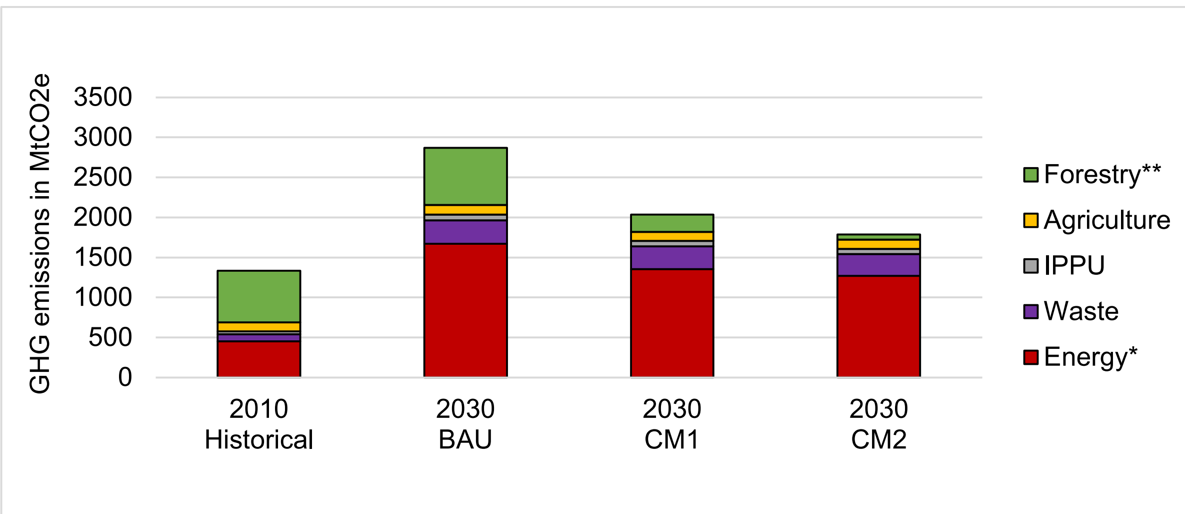

Fig. A1. Projected emissions in Indonesia's NDC in the Business as Usual Scenario (BAU), and the two conditional scenarios (CM1, CM2) in 2030 against 2010 historical emissions. *Including electricity, transport, heating, fugitive emissions, **including peat fires (Eckstein, Ordonez, & Wachsmuth, 2020).

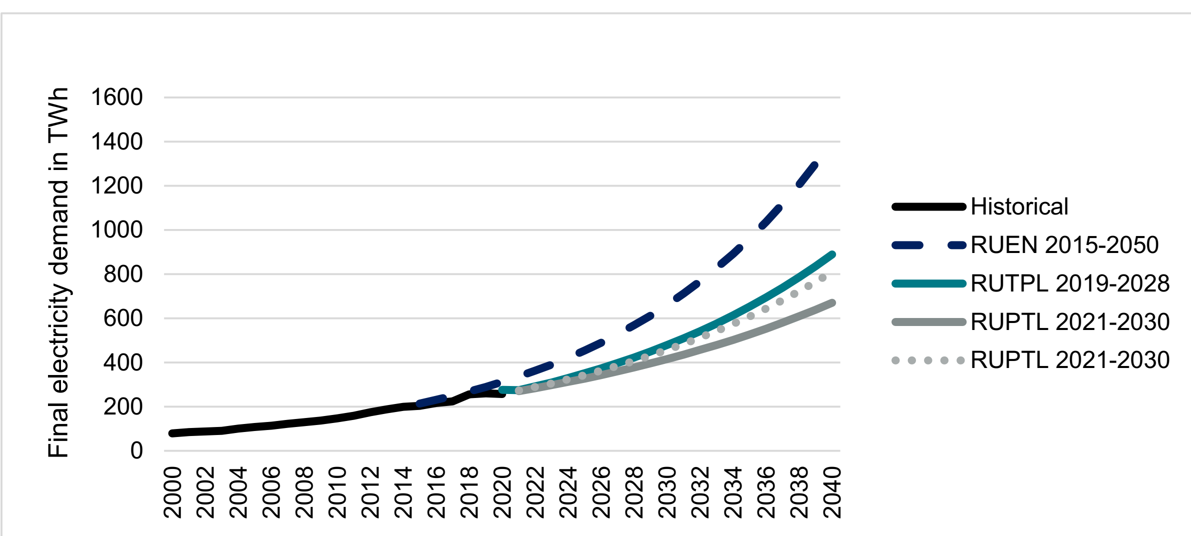

Fig. A2. Historical and projected total final electricity demand by using electricity demand growth rates as presented in RUPTL 2019–2028, RUPTL 2021–2030, and the energy masterplan RUEN. Own elaboration based on PLN, 2019; PLN, 2021; DEN, 2017b; ENERDATA, 2021).

Table A3
Power capacity in GW by technologies for official policy plans RUPTL 2019–2028, RUPTL 2021–2030 and RUEN, as well as for cost optimized scenarios for the years 2020, 2028, 2030, and 2040. Own elaboration based on scenario results and DEN, 2017a; PLN, 2019; PLN, 2021.

		Geo-thermal	Biomass	Solar PV	Wind	Large Hydro	Small hydro	Pumped Hydro Storage	Coal	CCGT	OCGT	Diesel	Total
RUPTL19	2020	2,3	0,1	0	0,1	5,9	0,5	0	39,8	14,3	8	6,5	77,5
	2028	6,6	0,9	1	0,9	10	1,7	3,5	59,3	20,4	9,6	6,6	120,5
RUPTL21	2020	2,3	0,1	0,2	0,1	5,9	0,5		39,8	14,2	8	6,1	77,2
	2028	4,6	1,3	4,5	0,7	9,5	1,6	2	53,6	18	9,9	6,2	111,9
	2030	5,7	1,9	4,8	0,8	10,9	1,6	4,2	53,6	18,1	10	6,2	117,8
RUEN	2020	3,1	1,1	0,9	0,6	5,6	1	0	37,6	15,1	6,6	5,4	77
	2028	8,4	5,1	10,4	4,1	20,3	3,5	0	66	22,9	10,9	7,6	159,2
	2030	9,3	7,8	14,2	7	22	3,8	0	74,6	25,6	12	8,3	184,6
	2040	13,4	17,6	29,6	17,5	30	5,4	0	122,8	42,2	19,8	13,6	311,9
Baseline	2020	2	0	0	0	5,4	0	0,1	32,5	11,3	6,3	6,4	64
	2028	1,8	0	0	0	5,4	0	6,6	58,6	8,9	5,7	5,9	92,9
	2030	5,8	0	0	0	5,4	0	7,8	60,4	8,9	5,1	5,6	99
	2040	16,5	0	0	0	5,4	0	21,2	85,7	5,3	2,4	3	139,5
RE cost reduction	2020	2	0	0	0	5,4	0	0,1	32,5	11,3	6,3	6,4	64
	2028	12	0,2	18,6	0	7,7	0	5	45,2	8,9	5,7	5,9	109,2
	2030	14,3	0,3	20,4	0	7,7	0	6,4	48,3	9	5,1	5,6	117,1
	2040	16,6	16,6	111,2	0	7,9	0	24	50,7	7,6	2,4	2,9	239,9
Combined policies	2020	2	1,9	0	0	5,4	0	0	32,2	11,2	6,3	6,4	65,4
	2028	15,1	15,3	55,6	0	7,7	2,9	1,6	29,5	8,8	5,7	5,9	148,1
	2030	16,3	15,3	73,9	0,9	7,7	2,9	4,9	29,2	8,8	5,1	5,6	170,6
	2040	16,6	23,8	139,1	0,9	8,4	2,9	12,3	22,9	32,2	2,4	3,4	264,9
CO_2 Price	2020	2,1	10,7	0	0	5,4	0	0	32,2	11,2	6,3	6,4	74,3
	2028	11,5	18,2	0	0	7,7	2,9	0,2	29,5	14,9	5,7	6,1	96,7
	2030	14,5	18,5	0	0	7,7	2,9	0,2	29,2	18,3	5,1	7,1	103,5
	2040	16,6	25,9	60	0	8,1	4,3	2,8	22,9	39,4	2,4	9,4	191,8

Appendix B. Modelling limitations

Of note, our model has several limitations. Cost reductions for RE in the absence of carbon pricing would require higher shares of solar PV in a power system with high shares of coal, and would demand much greater flexibility from the conventional power plant fleet. The applied modelling framework might lack the complexity needed to address the ramping up or down of conventional power plants, which may affect the modelling results, especially for scenarios with high RE shares towards the end of the modelling framework. The daily operation patterns should therefore only be seen as indicative. In any case, both demand side management measures and refurbishments of coal-power plants would be required to enable their ramping-up and down more flexibly (IRENA, 2021a). Such refurbishments, while theoretically possible, are subject to uncertainty as they might be costly or technically challenging. By focusing on the power sector, our model omits grid expansion constraints and transmission and distribution challenges in this regard. Another limitation is in assessing the sustainability of biomass use, as well as the technical and financial viability of pumped hydro storages. Adding substantial capacities of biomass comes along with the challenge of guaranteeing its sustainable management, in particular in a country where the bulk of emissions stem from the land use and forestry sector. Likewise, while large potential of pumped hydro storage is assumed to exist for Indonesia, its feasibility remains to be proved.

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